

Activity Content

Description

Application of Statistics

A quality control manager at a bottled water company collects the following data on the amount of water (in mL) in 15 randomly selected bottles labelled as 500 mL:

498, 502, 497, 503, 495, 499, 501, 504, 496, 500, 505, 506, 492, 493, 507

Tasks:

1. Find the Five-number summary (Minimum, Q1, Median, Q3, Maximum).
2. Calculate the Quartile Deviation (Semi-Interquartile Range).
3. Identify any outliers using the Interquartile Range (IQR) method.
4. Based on your findings, what can you say about the consistency of the bottling process?

Tips for writing posts:

- Read the prompt and instructions carefully.
- Identify the most important words or phrases in the prompt to help you stay on point.
- Ensure your post is free from plagiarism. When alluding to an idea or text/image that is not your own, add a citation that identifies the full name of the source, the date it was published, and any other citation element.
- Use a formal writing style.
- Double-check your post to ensure there are no spelling mistakes and grammatical issues.

Tips for responding to peer responses:

Responding to other participants is optional and does not affect your grades. However, responding to your peers' posts is recommended and encouraged.

While responding to your peers' posts

- Provide constructive comments.
- Provide reasons for your agreement or disagreement.

For constructing your arguments, we recommend that you use steelmanning, which implies that you find the best argument to test an opposing opinion. We do not recommend that you use strawman arguments. A strawman is a fallacious argument that distorts an opposing stance to make it easier to attack.

Click for more information about [Strawman Arguments](#).

Click for more information on [Knocking Down a Steel Man: How to Argue Better](#).